

Detailed Processing of the Stream (Sieve by Sieve)

Algorithm Overview

Given:

- **Submodular function:** $f(S) = |\bigcup_{e \in S} e|$
- **Desired subset size:** $k = 3$
- **Parameter:** $\epsilon = 0.3$
- **Stream of items (from right to left):**

$$e_1 = \{1\}, \quad e_2 = \{2, 3\}, \quad e_3 = \{3, 7\}, \quad e_4 = \{2, 4, 5\}, \quad e_5 = \{1, 2, 3\}, \quad e_6 = \{1, 5, 6, 8\}$$

Algorithm Steps

1. Initialization:

- Set $m := 0$.
- Define the threshold set $O = \{(1 + \epsilon)^i \mid i \in \mathbb{Z}\}$.
- For each $v \in O$, initialize $S_v := \emptyset$ (maintain only necessary v lazily).

2. Processing the Stream:

- For each element e_i in the stream (from $i = 1$ to n):

(a) **Update m :**

$$m := \max(m, f(\{e_i\})).$$

(b) **Define O_i :**

$$O_i = \{(1 + \epsilon)^j \mid m \leq (1 + \epsilon)^j \leq 2k \cdot m\}.$$

(c) **Delete obsolete solutions:** Remove all S_v where $v \notin O_i$.

(d) **Process each $v \in O_i$:**

- If $|S_v| < k$ and

$$\Delta f(e_i \mid S_v) \geq \frac{v/2 - f(S_v)}{k - |S_v|},$$

then update $S_v := S_v \cup \{e_i\}$.

(e) **Variables:**

- $\Delta f(e_i \mid S_v) = f(S_v \cup \{e_i\}) - f(S_v)$.
- $k - |S_v|$: Remaining capacity in S_v .

3. Output:

- Return $S = \arg \max_{v \in O_n} f(S_v)$.

Detailed Processing of the Stream

Step 1: Processing $e_1 = \{1\}$

1. Update m :

$$f(\{e_1\}) = |\{1\}| = 1$$

$$m = \max(0, 1) = 1$$

2. Define O_1 :

We need $v = (1 + \epsilon)^i$ such that $m \leq v \leq 2k \cdot m$.

Since $m = 1$ and $2k \cdot m = 6$, we require $v \in [1, 6]$.

Calculate v for integer $i \geq 0$:

$$\begin{aligned} i = 0 : \quad v &= (1.3)^0 = 1.0 \\ i = 1 : \quad v &= (1.3)^1 = 1.3 \\ i = 2 : \quad v &= (1.3)^2 = 1.69 \\ i = 3 : \quad v &= (1.3)^3 = 2.197 \\ i = 4 : \quad v &= (1.3)^4 = 2.8561 \\ i = 5 : \quad v &= (1.3)^5 = 3.7129 \\ i = 6 : \quad v &= (1.3)^6 = 4.8268 \end{aligned}$$

Therefore,

$$O_1 = \{1.0, 1.3, 1.69, 2.197, 2.8561, 3.7129, 4.8268\}.$$

3. Initialize $S_v = \emptyset$ for $v \in O_1$:

Total solutions constructed so far: **7**.

Initialize:

$$S_{1.0}, S_{1.3}, S_{1.69}, S_{2.197}, S_{2.8561}, S_{3.7129}, S_{4.8268}.$$

4. Process e_1 for each $v \in O_1$:

We will process each sieve S_v individually.

Sieve $v = 1.0$:

- Current S_v : $\emptyset$, $f(S_v) = 0$, $|S_v| = 0$.
- Remaining capacity: $k - |S_v| = 3$.
- Marginal gain: $\Delta f(e_1 \mid S_v) = f(\{e_1\}) - f(S_v) = 1 - 0 = 1$.
- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{1.0/2 - 0}{3} = \frac{0.5}{3} \approx 0.1667.$$

- Decision: Since $1 \geq 0.1667$, add e_1 to $S_{1.0}$.
- Updated $S_{1.0}$: $\{e_1\}$.

Sieve $v = 1.3$:

- Threshold: $\frac{1.3/2-0}{3} = \frac{0.65}{3} \approx 0.2167$.
- Decision: $1 \geq 0.2167$, add e_1 to $S_{1.3}$.
- Updated $S_{1.3}$: $\{e_1\}$.

Sieve $v = 1.69$:

- Threshold: $\frac{1.69/2-0}{3} \approx 0.2817$.
- Decision: $1 \geq 0.2817$, add e_1 to $S_{1.69}$.
- Updated $S_{1.69}$: $\{e_1\}$.

Sieve $v = 2.197$:

- Threshold: $\frac{2.197/2-0}{3} \approx 0.3662$.
- Decision: $1 \geq 0.3662$, add e_1 to $S_{2.197}$.
- Updated $S_{2.197}$: $\{e_1\}$.

Sieve $v = 2.8561$:

- Threshold: $\frac{2.8561/2-0}{3} \approx 0.4760$.
- Decision: $1 \geq 0.4760$, add e_1 to $S_{2.8561}$.
- Updated $S_{2.8561}$: $\{e_1\}$.

Sieve $v = 3.7129$:

- Threshold: $\frac{3.7129/2-0}{3} \approx 0.6188$.
- Decision: $1 \geq 0.6188$, add e_1 to $S_{3.7129}$.
- Updated $S_{3.7129}$: $\{e_1\}$.

Sieve $v = 4.8268$:

- Threshold: $\frac{4.8268/2-0}{3} \approx 0.8045$.
- Decision: $1 \geq 0.8045$, add e_1 to $S_{4.8268}$.
- Updated $S_{4.8268}$: $\{e_1\}$.

Summary After Step 1:

- All S_v now contain e_1 .
- Total solutions constructed: **7**.
- Total solutions discarded: **0**.

Step 2: Processing $e_2 = \{2, 3\}$

1. Update m :

$$\begin{aligned} f(\{e_2\}) &= |\{2, 3\}| = 2 \\ m &= \max(1, 2) = 2 \end{aligned}$$

2. Define O_2 :

Since $m = 2$ and $2k \cdot m = 12$, we require $v \in [2, 12]$.
Calculate v for integer i :

$$\begin{aligned} i = 3 : \quad v &= 2.197 \\ i = 4 : \quad v &= 2.8561 \\ i = 5 : \quad v &= 3.7129 \\ i = 6 : \quad v &= 4.8268 \\ i = 7 : \quad v &= 6.2748 \\ i = 8 : \quad v &= 8.1572 \\ i = 9 : \quad v &= 10.6044 \end{aligned}$$

Therefore,

$$O_2 = \{2.197, 2.8561, 3.7129, 4.8268, 6.2748, 8.1572, 10.6044\}.$$

3. Delete Obsolete S_v :

- Remove $S_{1.0}$, $S_{1.3}$, $S_{1.69}$ (since $v < m = 2$).
- Total solutions discarded: **3**.

4. Initialize New $S_v = \emptyset$ for New $v \in O_2$:

- New thresholds: 6.2748, 8.1572, 10.6044.
- Initialize $S_{6.2748}$, $S_{8.1572}$, $S_{10.6044}$.
- Additional solutions constructed: **3**.
- Total solutions constructed so far: $7 + 3 = \mathbf{10}$.

5. Process e_2 for each $v \in O_2$:

We will process each sieve S_v individually.

Sieve $v = 2.197$:

- Current S_v : $\{e_1\}$, $f(S_v) = 1$, $|S_v| = 1$.
- Remaining capacity: 2.
- Marginal gain:

$$\Delta f = f(\{e_1, e_2\}) - f(\{e_1\}) = |\{1, 2, 3\}| - 1 = 3 - 1 = 2.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{2.197/2 - 1}{2} \approx 0.0493.$$

- Decision: $2 \geq 0.0493$, add e_2 to $S_{2.197}$.
- Updated $S_{2.197}$: $\{e_1, e_2\}$.

Sieve $v = 2.8561$:

- Threshold: $\frac{2.8561/2-1}{2} \approx 0.2140$.
- Decision: $2 \geq 0.2140$, add e_2 to $S_{2.8561}$.
- Updated $S_{2.8561}$: $\{e_1, e_2\}$.

Sieve $v = 3.7129$:

- Threshold: $\frac{3.7129/2-1}{2} \approx 0.4282$.
- Decision: $2 \geq 0.4282$, add e_2 to $S_{3.7129}$.
- Updated $S_{3.7129}$: $\{e_1, e_2\}$.

Sieve $v = 4.8268$:

- Threshold: $\frac{4.8268/2-1}{2} \approx 0.7067$.
- Decision: $2 \geq 0.7067$, add e_2 to $S_{4.8268}$.
- Updated $S_{4.8268}$: $\{e_1, e_2\}$.

Sieve $v = 6.2748$:

- Current S_v : $\emptyset$, $f(S_v) = 0$, $|S_v| = 0$.
- Remaining capacity: 3.
- Marginal gain: $\Delta f = f(\{e_2\}) - 0 = 2$.
- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{6.2748/2 - 0}{3} = \frac{3.1374}{3} \approx 1.0458.$$

- Decision: $2 \geq 1.0458$, add e_2 to $S_{6.2748}$.
- Updated $S_{6.2748}$: $\{e_2\}$.

Sieve $v = 8.1572$:

- Threshold: $\frac{8.1572/2 - 0}{3} \approx 1.3595$.
- Decision: $2 \geq 1.3595$, add e_2 to $S_{8.1572}$.
- Updated $S_{8.1572}$: $\{e_2\}$.

Sieve $v = 10.6044$:

- Threshold: $\frac{10.6044/2 - 0}{3} \approx 1.7674$.
- Decision: $2 \geq 1.7674$, add e_2 to $S_{10.6044}$.
- Updated $S_{10.6044}$: $\{e_2\}$.

Summary After Step 2:

- Total solutions constructed: **10**.
- Total solutions discarded: **3**.
- Updated S_v :
 - $S_{2.197}$, $S_{2.8561}$, $S_{3.7129}$, $S_{4.8268}$ contain $\{e_1, e_2\}$.
 - $S_{6.2748}$, $S_{8.1572}$, $S_{10.6044}$ contain $\{e_2\}$.

Step 3: Processing $e_3 = \{3, 7\}$

1. Update m :

$$f(\{e_3\}) = |\{3, 7\}| = 2$$
$$m = \max(2, 2) = 2$$

2. Define O_3 :

Since $m = 2$ and $2k \cdot m = 12$, we require $v \in [2, 12]$.

Therefore, O_3 remains the same as O_2 :

$$O_3 = \{2.197, 2.8561, 3.7129, 4.8268, 6.2748, 8.1572, 10.6044\}.$$

3. Delete Obsolete S_v :

- No S_v are deleted since $O_3 = O_2$.

4. Initialize New $S_v = \emptyset$ for New $v \in O_3$:

- No new S_v are initialized.

5. Process e_3 for each $v \in O_3$:

We will process each sieve S_v individually.

Sieve $v = 2.197$:

- Current S_v : $\{e_1, e_2\}$, $f(S_v) = |\{1, 2, 3\}| = 3$, $|S_v| = 2$.
- Remaining capacity: $k - |S_v| = 1$.
- Marginal gain:

$$f(S_v \cup \{e_3\}) = |\{1, 2, 3, 7\}| = 4,$$
$$\Delta f(e_3 | S_v) = 4 - 3 = 1.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{2.197/2 - 3}{1} = \frac{1.0985 - 3}{1} = -1.9015.$$

- Decision: Since $1 \geq -1.9015$, add e_3 to $S_{2.197}$.
- Updated $S_{2.197}$: $\{e_1, e_2, e_3\}$, $|S_v| = 3$.

Sieve $v = 2.8561$:

- Threshold:

$$\text{Threshold} = \frac{2.8561/2 - 3}{1} = \frac{1.42805 - 3}{1} = -1.57195.$$

- Decision: $1 \geq -1.57195$, add e_3 to $S_{2.8561}$.
- Updated $S_{2.8561}$: $\{e_1, e_2, e_3\}$, $|S_v| = 3$.

Sieve $v = 3.7129$:

- Threshold:

$$\text{Threshold} = \frac{3.7129/2 - 3}{1} = \frac{1.85645 - 3}{1} = -1.14355.$$

- Decision: $1 \geq -1.14355$, add e_3 to $S_{3.7129}$.
- Updated $S_{3.7129}$: $\{e_1, e_2, e_3\}$, $|S_v| = 3$.

Sieve $v = 4.8268$:

- Threshold:

$$\text{Threshold} = \frac{4.8268/2 - 3}{1} = \frac{2.4134 - 3}{1} = -0.5866.$$

- Decision: $1 \geq -0.5866$, add e_3 to $S_{4.8268}$.
- Updated $S_{4.8268}$: $\{e_1, e_2, e_3\}$, $|S_v| = 3$.

Sieve $v = 6.2748$:

- Current S_v : $\{e_2\}$, $f(S_v) = 2$, $|S_v| = 1$.
- Remaining capacity: 2.
- Marginal gain:

$$f(S_v \cup \{e_3\}) = |\{2, 3, 7\}| = 3, \quad \Delta f = 3 - 2 = 1.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{6.2748/2 - 2}{2} = \frac{3.1374 - 2}{2} = 0.5687.$$

- Decision: $1 \geq 0.5687$, add e_3 to $S_{6.2748}$.
- Updated $S_{6.2748}$: $\{e_2, e_3\}$, $|S_v| = 2$.

Sieve $v = 8.1572$:

- Threshold:

$$\text{Threshold} = \frac{8.1572/2 - 2}{2} = \frac{4.0786 - 2}{2} = 1.0393.$$

- Decision: $1 \not\geq 1.0393$, do not add e_3 to $S_{8.1572}$.
- Updated $S_{8.1572}$: Remains $\{e_2\}$, $|S_v| = 1$.

Sieve $v = 10.6044$:

- Threshold:

$$\text{Threshold} = \frac{10.6044/2 - 2}{2} = \frac{5.3022 - 2}{2} = 1.6511.$$

- Decision: $1 \not\geq 1.6511$, do not add e_3 to $S_{10.6044}$.
- Updated $S_{10.6044}$: Remains $\{e_2\}$, $|S_v| = 1$.

Summary After Step 3:

- $S_{2.197}$, $S_{2.8561}$, $S_{3.7129}$, $S_{4.8268}$ now have $\{e_1, e_2, e_3\}$ (full).
- $S_{6.2748}$ has $\{e_2, e_3\}$, $|S_v| = 2$.
- $S_{8.1572}$, $S_{10.6044}$ remain with $\{e_2\}$, $|S_v| = 1$.

Step 4: Processing $e_4 = \{2, 4, 5\}$

1. Update m :

$$\begin{aligned} f(\{e_4\}) &= |\{2, 4, 5\}| = 3 \\ m &= \max(2, 3) = 3 \end{aligned}$$

2. Define O_4 :

Since $m = 3$ and $2k \cdot m = 18$, we require $v \in [3, 18]$.
Calculating v for integer i :

$$\begin{aligned} i = 5 : \quad v &= 3.7129 \\ i = 6 : \quad v &= 4.8268 \\ i = 7 : \quad v &= 6.2748 \\ i = 8 : \quad v &= 8.1572 \\ i = 9 : \quad v &= 10.6044 \\ i = 10 : \quad v &= 13.7857 \\ i = 11 : \quad v &= 17.9215 \end{aligned}$$

Therefore,

$$O_4 = \{3.7129, 4.8268, 6.2748, 8.1572, 10.6044, 13.7857, 17.9215\}.$$

3. Delete Obsolete S_v :

- Remove $S_{2.197}$, $S_{2.8561}$ (since $v < m = 3$).
- Total solutions discarded: Previously 3, now $3 + 2 = 5$.

4. Initialize New $S_v = \emptyset$ for New $v \in O_4$:

- New thresholds: 13.7857, 17.9215.
- Initialize $S_{13.7857}$, $S_{17.9215}$.
- Additional solutions constructed: **2**.
- Total solutions constructed so far: $10 + 2 = \mathbf{12}$.

5. Process e_4 for each $v \in O_4$:

Sieve $v = 3.7129$:

- Current S_v : $\{e_1, e_2, e_3\}$, $|S_v| = 3$ (full).
- Decision: Cannot add e_4 , S_v is full.

Sieve $v = 4.8268$:

- Current S_v : $\{e_1, e_2, e_3\}$, $|S_v| = 3$ (full).
- Decision: Cannot add e_4 , S_v is full.

Sieve $v = 6.2748$:

- Current S_v : $\{e_2, e_3\}$, $f(S_v) = 3$, $|S_v| = 2$.
- Remaining capacity: 1.
- Marginal gain:

$$f(S_v \cup \{e_4\}) = |\{2, 3, 4, 5, 7\}| = 5, \quad \Delta f = 5 - 3 = 2.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{6.2748/2 - 3}{1} = 0.1374.$$

- Decision: $2 \geq 0.1374$, add e_4 to $S_{6.2748}$.
- Updated $S_{6.2748}$: $\{e_2, e_3, e_4\}$, $|S_v| = 3$.

Sieve $v = 8.1572$:

- Current S_v : $\{e_2\}$, $f(S_v) = 2$, $|S_v| = 1$.
- Remaining capacity: 2.
- Marginal gain:

$$f(S_v \cup \{e_4\}) = |\{2, 3, 4, 5\}| = 4, \quad \Delta f = 4 - 2 = 2.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{8.1572/2 - 2}{2} = 1.0393.$$

- Decision: $2 \geq 1.0393$, add e_4 to $S_{8.1572}$.
- Updated $S_{8.1572}$: $\{e_2, e_4\}$, $|S_v| = 2$.

Sieve $v = 10.6044$:

- Threshold:

$$\text{Threshold} = \frac{10.6044/2 - 2}{2} = 1.6511.$$

- Decision: $2 \geq 1.6511$, add e_4 to $S_{10.6044}$.
- Updated $S_{10.6044}$: $\{e_2, e_4\}$, $|S_v| = 2$.

Sieve $v = 13.7857$:

- Current S_v : $\emptyset$, $f(S_v) = 0$, $|S_v| = 0$.
- Remaining capacity: 3.
- Marginal gain: $\Delta f = f(\{e_4\}) - 0 = 3$.
- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{13.7857/2 - 0}{3} = \frac{6.89285}{3} \approx 2.2976.$$

- Decision: $3 \geq 2.2976$, add e_4 to $S_{13.7857}$.
- Updated $S_{13.7857}$: $\{e_4\}$, $|S_v| = 1$.

Sieve $v = 17.9215$:

- Threshold:

$$\text{Threshold} = \frac{17.9215/2 - 0}{3} = \frac{8.96075}{3} \approx 2.9869.$$

- Decision: $3 \geq 2.9869$, add e_4 to $S_{17.9215}$.
- Updated $S_{17.9215}$: $\{e_4\}$, $|S_v| = 1$.

Summary After Step 4:

- Total solutions constructed: **12**.
- Total solutions discarded: **5**.
- Updated S_v :
 - $S_{3.7129}$, $S_{4.8268}$, $S_{6.2748}$ have $|S_v| = 3$.
 - $S_{8.1572}$, $S_{10.6044}$ have $|S_v| = 2$.
 - $S_{13.7857}$, $S_{17.9215}$ have $|S_v| = 1$.

Step 5: Processing $e_5 = \{1, 2, 3\}$

1. Update m :

$$\begin{aligned} f(\{e_5\}) &= |\{1, 2, 3\}| = 3 \\ m &= \max(3, 3) = 3 \end{aligned}$$

2. Define O_5 :

Since $m = 3$ and $2k \cdot m = 18$, we require $v \in [3, 18]$.

Therefore, O_5 remains the same as O_4 :

$$O_5 = \{3.7129, 4.8268, 6.2748, 8.1572, 10.6044, 13.7857, 17.9215\}.$$

3. Delete Obsolete S_v :

- No S_v are deleted since $O_5 = O_4$.

4. Initialize New $S_v = \emptyset$ for New $v \in O_5$:

- No new S_v are initialized.

5. Process e_5 for each $v \in O_5$:

We will process each sieve S_v individually.

Sieve $v = 3.7129$:

- Current S_v : $\{e_1, e_2, e_3\}$, $|S_v| = 3$ (full).
- Decision: Cannot add e_5 , S_v is full.

Sieve $v = 4.8268$:

- Current S_v : $\{e_1, e_2, e_3\}$, $|S_v| = 3$ (full).
- Decision: Cannot add e_5 , S_v is full.

Sieve $v = 6.2748$:

- Current S_v : $\{e_2, e_3, e_4\}$, $|S_v| = 3$ (full).
- Decision: Cannot add e_5 , S_v is full.

Sieve $v = 8.1572$:

- Current S_v : $\{e_2, e_4\}$, $f(S_v) = |\{2, 3, 4, 5\}| = 4$, $|S_v| = 2$.
- Remaining capacity: 1.
- Marginal gain:

$$\begin{aligned} f(S_v \cup \{e_5\}) &= |\{1, 2, 3, 4, 5\}| = 5, \\ \Delta f(e_5 \mid S_v) &= 5 - 4 = 1. \end{aligned}$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{8.1572/2 - 4}{1} = \frac{4.0786 - 4}{1} = 0.0786.$$

- Decision: Since $1 \geq 0.0786$, add e_5 to $S_{8.1572}$.
- Updated $S_{8.1572}$: $\{e_2, e_4, e_5\}, |S_v| = 3$.

Sieve $v = 10.6044$:

- Current S_v : $\{e_2, e_4\}, f(S_v) = 4, |S_v| = 2$.
- Remaining capacity: 1.
- Marginal gain:

$$f(S_v \cup \{e_5\}) = |\{1, 2, 3, 4, 5\}| = 5, \quad \Delta f = 5 - 4 = 1.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{10.6044/2 - 4}{1} = \frac{5.3022 - 4}{1} = 1.3022.$$

- Decision: Since $1 \not\geq 1.3022$, do not add e_5 to $S_{10.6044}$.
- Updated $S_{10.6044}$: Remains $\{e_2, e_4\}, |S_v| = 2$.

Sieve $v = 13.7857$:

- Current S_v : $\{e_4\}, f(S_v) = 3, |S_v| = 1$.
- Remaining capacity: 2.
- Marginal gain:

$$f(S_v \cup \{e_5\}) = |\{1, 2, 3, 4, 5\}| = 5, \\ \Delta f(e_5 | S_v) = 5 - 3 = 2.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{13.7857/2 - 3}{2} = \frac{6.89285 - 3}{2} = \frac{3.89285}{2} = 1.9464.$$

- Decision: Since $2 \geq 1.9464$, add e_5 to $S_{13.7857}$.
- Updated $S_{13.7857}$: $\{e_4, e_5\}, |S_v| = 2$.

Sieve $v = 17.9215$:

- Current S_v : $\{e_4\}$, $f(S_v) = 3$, $|S_v| = 1$.
- Remaining capacity: 2.
- Marginal gain:

$$f(S_v \cup \{e_5\}) = 5 - 3 = 2.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{17.9215/2 - 3}{2} = \frac{8.96075 - 3}{2} = \frac{5.96075}{2} = 2.9804.$$

- Decision: Since $2 \not\geq 2.9804$, do not add e_5 to $S_{17.9215}$.
- Updated $S_{17.9215}$: Remains $\{e_4\}$, $|S_v| = 1$.

Summary After Step 5:

- Total solutions constructed: Remains at **12**.
- Total solutions discarded: Remains at **5**.
- Updated S_v :
 - $S_{8.1572}$ is now full ($|S_v| = 3$).
 - $S_{13.7857}$ has $|S_v| = 2$.
 - Other S_v remain unchanged.

Step 6: Processing $e_6 = \{1, 5, 6, 8\}$

1. Update m :

$$f(\{e_6\}) = |\{1, 5, 6, 8\}| = 4$$

$$m = \max(3, 4) = 4$$

2. Define O_6 :

Since $m = 4$ and $2k \cdot m = 24$, we require $v \in [4, 24]$.

Calculating v for integer i :

$$i = 12 : \quad v = (1.3)^{12} \approx 23.2979$$

Therefore,

$$O_6 = \{4.8268, 6.2748, 8.1572, 10.6044, 13.7857, 17.9215, 23.2979\}.$$

3. Delete Obsolete S_v :

- Remove $S_{3.7129}$ (since $v = 3.7129 < m = 4$).
- Total solutions discarded: **6**.
- 4. Initialize New $S_v = \emptyset$ for New $v \in O_6$:**
- New threshold: $v = 23.2979$.
- Initialize $S_{23.2979}$.
- Additional solutions constructed: **1**.
- Total solutions constructed so far: $12 + 1 = \mathbf{13}$.

5. Process e_6 for each $v \in O_6$:

We will process each sieve S_v individually.

Sieve $v = 4.8268$:

- Current S_v : $\{e_1, e_2, e_3\}$, $|S_v| = 3$ (full).
- Decision: Cannot add e_6 , S_v is full.

Sieve $v = 6.2748$:

- Current S_v : $\{e_2, e_3, e_4\}$, $|S_v| = 3$ (full).
- Decision: Cannot add e_6 , S_v is full.

Sieve $v = 8.1572$:

- Current S_v : $\{e_2, e_4, e_5\}$, $|S_v| = 3$ (full).
- Decision: Cannot add e_6 , S_v is full.

Sieve $v = 10.6044$:

- Current S_v : $\{e_2, e_4\}$, $f(S_v) = 4$, $|S_v| = 2$.
- Remaining capacity: 1.
- Marginal gain:

$$f(S_v \cup \{e_6\}) = |\{1, 2, 3, 4, 5, 6, 8\}| = 7,$$

$$\Delta f = 7 - 4 = 3.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{10.6044/2 - 4}{1} = \frac{5.3022 - 4}{1} = 1.3022.$$

- Decision: Since $3 \geq 1.3022$, add e_6 to $S_{10.6044}$.
- Updated $S_{10.6044}$: $\{e_2, e_4, e_6\}$, $|S_v| = 3$.

Sieve $v = 13.7857$:

- Current S_v : $\{e_4, e_5\}$, $f(S_v) = 5$, $|S_v| = 2$.
- Remaining capacity: 1.
- Marginal gain:

$$f(S_v \cup \{e_6\}) = |\{1, 2, 3, 4, 5, 6, 8\}| = 7,$$

$$\Delta f = 7 - 5 = 2.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{13.7857/2 - 5}{1} = \frac{6.89285 - 5}{1} = 1.89285.$$

- Decision: Since $2 \geq 1.89285$, add e_6 to $S_{13.7857}$.
- Updated $S_{13.7857}$: $\{e_4, e_5, e_6\}$, $|S_v| = 3$.

Sieve $v = 17.9215$:

- Current S_v : $\{e_4\}$, $f(S_v) = 3$, $|S_v| = 1$.
- Remaining capacity: 2.
- Marginal gain:

$$f(S_v \cup \{e_6\}) = |\{1, 2, 4, 5, 6, 8\}| = 6,$$

$$\Delta f = 6 - 3 = 3.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{17.9215/2 - 3}{2} = \frac{8.96075 - 3}{2} = \frac{5.96075}{2} = 2.9804.$$

- Decision: Since $3 \geq 2.9804$, add e_6 to $S_{17.9215}$.
- Updated $S_{17.9215}$: $\{e_4, e_6\}$, $|S_v| = 2$.

Sieve $v = 23.2979$:

- Current S_v : $\emptyset$, $f(S_v) = 0$, $|S_v| = 0$.
- Remaining capacity: 3.
- Marginal gain:

$$f(S_v \cup \{e_6\}) = |\{1, 5, 6, 8\}| = 4, \quad \Delta f = 4 - 0 = 4.$$

- Threshold:

$$\text{Threshold} = \frac{v/2 - f(S_v)}{k - |S_v|} = \frac{23.2979/2 - 0}{3} = \frac{11.64895}{3} \approx 3.8830.$$

- Decision: Since $4 \geq 3.8830$, add e_6 to $S_{23.2979}$.
- Updated $S_{23.2979}$: $\{e_6\}$, $|S_v| = 1$.

Summary After Step 6:

- Total solutions constructed: **13**.
- Total solutions discarded: **6**.
- Updated S_v :

- $S_{10.6044}$ is now full ($|S_v| = 3$).
- $S_{13.7857}$ is now full ($|S_v| = 3$).
- $S_{17.9215}$ has $|S_v| = 2$.
- $S_{23.2979}$ has $|S_v| = 1$.
- Other S_v remain unchanged.

Final Result:

Total Number of Solutions Constructed (Including Discarded): 13

These are:

- **From Step 1:** 7 solutions

$$- S_{1.0}, S_{1.3}, S_{1.69}, S_{2.197}, S_{2.8561}, S_{3.7129}, S_{4.8268}$$

- **From Step 2:** 3 new solutions

$$- S_{6.2748}, S_{8.1572}, S_{10.6044}$$

- **From Step 4:** 2 new solutions

$$- S_{13.7857}, S_{17.9215}$$

- **From Step 6:** 1 new solution

$$- S_{23.2979}$$

Total Solutions Discarded: 6

- $S_{1.0}, S_{1.3}, S_{1.69}, S_{2.197}, S_{2.8561}, S_{3.7129}$